Quantification: 872_wt_hp_z04
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	13
interior lacuna count	10
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 10)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2101.33	2144	927.448	px^2
major_axis_length_px	lacuna shape	80.9142	76.9066	19.4155	px
minor_axis_length_px	lacuna shape	35.0904	34.0354	9.2461	px
aspect_ratio	lacuna shape	2.3403	2.5083	0.4528	unitless
eccentricity	lacuna shape	0.8904	0.9171	0.0524	unitless
solidity	lacuna shape	0.7707	0.7683	0.0678	unitless
roots_count	headline	15	15	4.8074	count
ring_length_r30_px	headline	694.4	762	190.362	px
ring_length_r60_px	reported	1621.5	1752.5	392.405	px
owned_length_px	ownership-dependent	3389.98	3025.67	1369.53	px
edge_count	ownership-dependent	219.8	199.5	94.2288	count
mean_edge_length_px	ownership-dependent	15.6519	15.8886	1.6758	px
perimeter_px	normalised	254.63	264.764	86.6163	px
ring_area_r30_px2	normalised	9022	9101.5	2028.2	px^2
ring_area_r60_px2	normalised	22120.7	22141.5	4248.88	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0769457	0.0802137	0.0112555	px^-1
ring_density_r60	normalised	0.0738819	0.0744297	0.01391	px^-1
roots_per_100px_perimeter	normalised	6.1029	6.1795	1.4389	per 100 px
ring_attached_length_r30_px	network v2	664.4	712.5	195.337	px
ring_attached_length_r60_px	network v2	1508.9	1697.5	456.382	px
ring_length_w_r30_px	network v2	796.383	875.26	218.501	px
ring_length_w_r60_px	network v2	1849.05	2000.51	461.215	px
sholl_crossings_r10	network v2	15	16	4.4969	count
sholl_crossings_r20	network v2	16.3	16.5	4.5228	count
sholl_crossings_r30	network v2	21	21	6.5149	count
loop_count	loops, report only	35.4	34	24.4595	count
loop_edge_share	loops, report only	0.5496	0.6148	0.1482	share
ring_width_mean_r30_px	width	7.2029	7.2026	0.3875	px

Per field
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measure	value	units
analysed_area_px2	877862	px^2
total_skeleton_length_px	36616	px
canalicular_length_density_per_px	0.0417104	px^-1
junction_count	1331	count
junction_density_per_px2	0.00151618	px^-2
lacuna_count	13	count

lacunae_per_px2	1.481e-05	px^-2
skeleton_component_count	111	count
mean_component_length_px	329.874	px
median_component_length_px	12	px
field_length_density_w_per_px	0.0473629	px^-1
field_density_without_flagged_per_px	0.0368141	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	43	px
unattached_component_count	3	count
canalicular_length_density_attached_per_px	0.0417104	px^-1
field_length_density_with_unattached_per_px	0.0417594	px^-1
excluded_area_px2	143103	px^2
analysed_field_area_px2	905473	px^2
loop_count	493	count
loop_edge_share	0.6155	share
edge_count	2762	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7432	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	37433	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	3025	100.943	44.409	2.273	0.898	0.65	-1.4641	211.119	126.158	False
2	2144	87.4647	34.0354	2.5698	0.9212	0.7528	-1.2557	226.871	777.66	False
3	2841	103.213	37.4578	2.7554	0.9318	0.8351	-1.3658	276.205	420.61	False
4	3125	68.6713	60.8024	1.1294	0.4648	0.909	0.622	332.681	995.111	True
5	3954	114.494	48.2385	2.3735	0.9069	0.7517	-1.4049	393.248	55.5005	True
6	3573	94.8523	52.0807	1.8213	0.8358	0.8284	-1.4851	395.903	280.417	False
7	1409	76.4593	28.6406	2.6696	0.9272	0.7683	-1.5183	392.216	831.581	False
8	2226	76.9066	39.7827	1.9332	0.8558	0.7987	-1.558	506.567	185.868	False
9	1503	75.7312	26.0285	2.9095	0.9391	0.8628	-1.3051	568.045	702.943	False
10	1123	61.8357	30.8107	2.007	0.867	0.6785	-1.3397	588.985	20.1576	True
11	628	38.8866	23.9678	1.6225	0.7875	0.7235	-1.4785	619.033	332.451	False
12	1563	73.771	29.411	2.5083	0.9171	0.717	-1.5252	651.639	225.151	False
13								264.441	629.968	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	23	955	1991	2872.15	181	15.8682	7.1991
2	19	770	1744	2899.12	179	16.1962	7.1819
3	14	754	1880	3468.16	218	15.909	7.206
4	13	390	891	1463.58	97	15.0885	7.5469

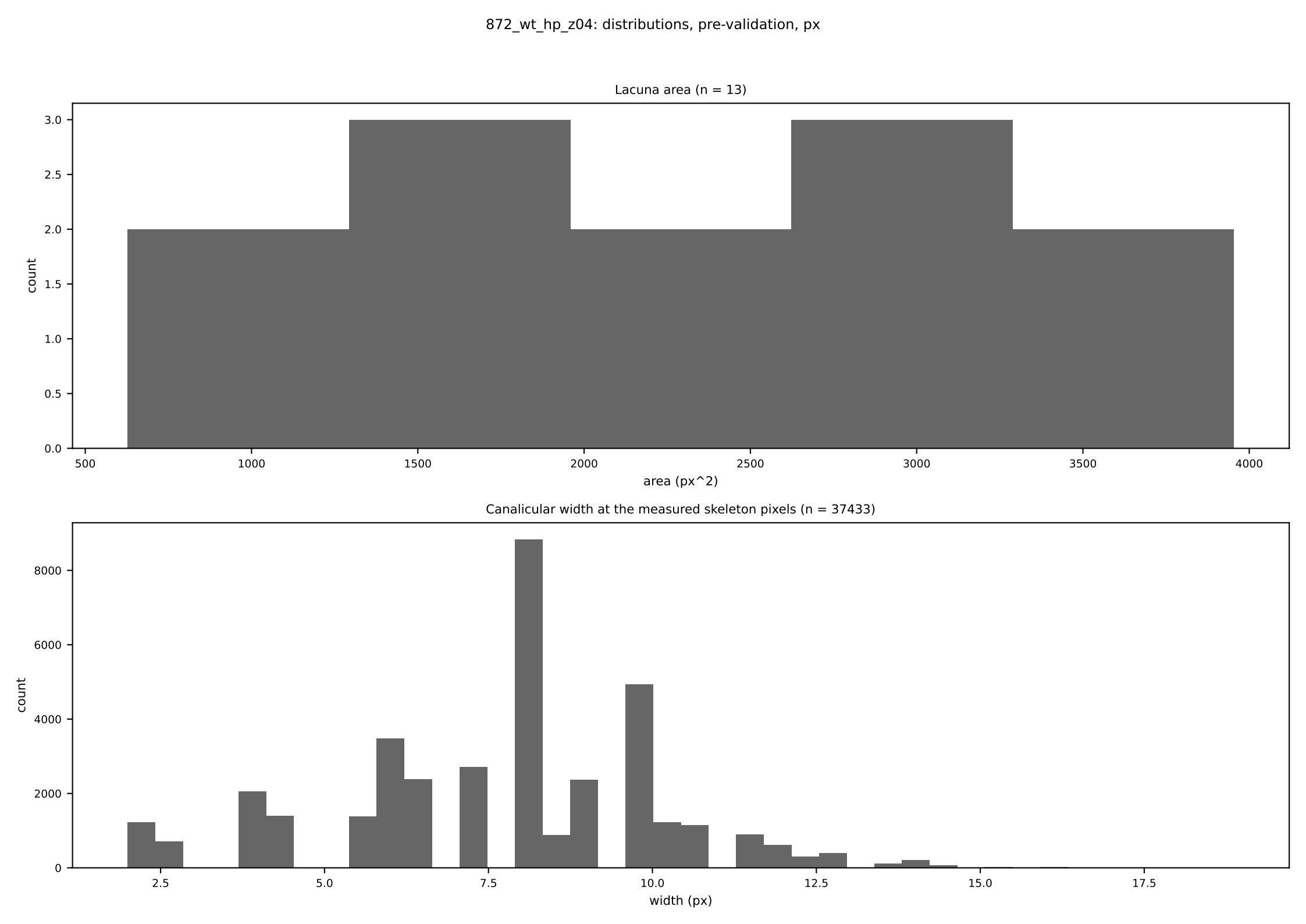
5	17	612	1367	1571.92	106	14.8295	7.266
6	16	921	2187	4853.03	318	15.2611	7.767
7	11	464	1199	2383.31	136	17.5244	6.5749
8	20	794	1881	2130.37	174	12.2435	7.3004
9	9	507	1064	2000.2	110	18.1836	6.8661
10	12	482	1240	2288.55	153	14.9578	6.8827
11	12	528	1281	6482.65	434	14.937	6.8215
12	17	802	1761	3152.23	223	14.1356	7.3103
13	9	449	1227	3658.62	225	16.2605	7.8017

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	425.345	11873	28650	1	1	0.0804346	0.0694939	5.4074
2	273.321	9846	24205	1	1	0.0782044	0.0720512	6.9515
3	288.978	10581	26042	1	1	0.0712598	0.0721911	4.8447
4	236.551	6707	15888	0.711	0.6585	0.0581482	0.0560801	5.4956
5	432.996	10444	22425	0.8508	0.7745	0.0585982	0.0609587	3.9261
6	315.191	11229	25289	1	1	0.0820198	0.0864803	5.0763
7	256.208	8359	21924	1	1	0.055509	0.0546889	4.2934
8	273.764	9314	21493	1	1	0.085248	0.0875169	7.3055
9	204.445	8382	22359	1	1	0.0604868	0.0475871	4.4022
10	216.752	5820	13951	0.7153	0.655	0.0828179	0.0888825	5.5363
11	148.61	6134	15204	1	1	0.0860776	0.0842541	8.0748
12	243.386	8889	20037	1	1	0.0902239	0.0878874	6.9848
13	117.054	5613	16004	1	1	0.0799929	0.0766683	7.6888

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	930	1748	1101.89	2270.4	22	18	26
2	693	1661	886.867	1995.75	17	16	26
3	732	1812	863.654	2153.24	18	19	22
4	354	721	435.948	1014.26	8	12	12
5	595	1259	712.097	1568.12	14	14	14
6	901	2184	1058.47	2524.25	20	25	31
7	412	1020	531.274	1353.7	10	14	14
8	790	1864	898.512	2154.37	16	17	27
9	459	740	574.132	1183.22	11	8	15
10	475	1239	544.676	1418.12	12	12	14
11	495	1176	598.86	1449.87	11	14	20
12	784	1734	919.583	2005.28	16	19	18
13	448	1150	530.588	1400.36	9	13	11


872_wt_hp_z04: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 37433)
count
width (px)